THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A crystal of potassium permanganate is dropped into water in two beakers, one hot and one cold. The colour spreads faster in the hot beaker because: 1

(a) water gets heavier
(b) crystal dissolves into gas
(c) particles move faster at higher temperature
(d) cold water is thicker

2. Steam at 100 degrees Celsius has more energy than water at 100 degrees Celsius because it has: 1

(a) latent heat of vaporisation **(b)** more mass
(c) lower temperature **(d)** more particles

3. A gas at high pressure and low temperature is more likely to become: 1

(a) solid always **(b)** vapour only **(c)** plasma **(d)** liquid

4. When 10 mL of salt is added to 100 mL of water, the total volume is less than 110 mL. This is because: 1

(a) salt becomes gas
(b) water evaporates entirely
(c) particles of salt fit in the gaps between water particles
(d) salt disappears

5. A gas can be turned into a liquid if it is cooled below its: 1

(a) melting point only **(b)** boiling point (condensation point)
(c) room temperature **(d)** freezing point only

6. Rubbing a drop of spirit on the palm gives a cooling feeling because spirit: 1

(a) evaporates quickly **(b)** burns
(c) is heavy **(d)** is cold

7. Which of the following sets are in the right order of increasing space between particles? 1

(a) gas, solid, liquid **(b)** liquid, solid, gas
(c) solid, liquid, gas **(d)** gas, liquid, solid

8. Water at 4 degrees Celsius and 90 degrees Celsius: the faster-moving particles are in: 1

(a) water at 90 degrees Celsius **(b)** water at 4 degrees Celsius
(c) equal **(d)** none

9. Why does a closed container of water not appear to lose water? 1

(a) water is solid
(b) vapour cannot escape; condensation balances evaporation
(c) pressure is low
(d) no evaporation occurs

10. The smell of incense sticks reaches the farthest room slowly because: 1

(a) diffusion in air takes time **(b)** air is a solid
(c) incense is a liquid **(d)** incense does not diffuse

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the volume of a mixture of alcohol and water appear less than the sum of their volumes? 2

12. State the effect of temperature on the speed of particles and diffusion. 2

13. Differentiate between vapour and gas. 2

14. Why does a drop of water on a leaf form a bead? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why solids retain their shape, liquids flow and gases spread using particle ideas. 3

16. Explain the change in arrangement of particles when water vapour changes to ice. 3

17. Discuss the effect of area of surface on the rate of evaporation with an example. 3

OR

Explain why the rate of diffusion in gases is more than in liquids. 3

18. Explain why the humid climate on the coast makes sweat evaporate slowly and makes us feel uncomfortable. 3

OR

What are the differences between melting and dissolving? Give two points. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an experiment to show that particles of matter are continuously moving and have spaces between them, using KMnO4 and water. 5

OR

Describe how pressure and temperature can be used to liquefy gases and give three uses. 5

20. Explain the heating curve of water from -10 degrees Celsius to 110 degrees Celsius with the state and temperature in each part. 5

OR

A student kept equal amounts of water in a bowl, a glass and a saucer in the sun. Predict the order of drying and explain. 5

END OF QUESTION PAPER Total: 40 Marks